

Deep Learning for AI-Generated Image Detection: A Review of CNN, Vision Transformer, and Explainable Approaches

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ABSTRACT

Text-to-image generation is now cheap and publicly available, and its output is increasingly hard to separate from photographs by eye. The consequences — fabricated evidence, misinformation, impersonation, and contamination of image corpora scraped from the web — arrive faster than manual inspection can handle, so automated detection has become a working requirement. This paper reviews deep learning approaches to detecting AI-generated still images. Following a protocol fixed before collection, six databases were searched with eight query seeds; 48 primary studies published between 2019 and 2026 were read in full and entered into a structured extraction table, with video and audio deepfake detection excluded. The surveyed work is organised by the origin of the evidence a detector uses: handcrafted and frequency-domain forensics, convolutional networks trained by transfer learning, transformer and attention-based models, detectors built on frozen vision-language representations, and explainability-oriented work. Datasets, benchmarks and evaluation metrics are compared in the same way. The central finding is that generalisation across generators is a structural problem, because much of what a detector learns belongs to a particular generator's design: reported accuracy falls from 99.9% on a training generator to 54.9% on an unseen one, and from 92.77% on a standard benchmark to 65.77% on realistic in-the-wild images. The strongest cross-generator results share frozen general-purpose representations, derived inputs, or diverse training data. Cross-generator evaluation is reported by 36 of the 48 studies and explanation analysis by four, but almost never together — the gap that motivates the study proposed here.

1. Introduction

Generative adversarial networks (GANs), proposed in 2014, were the first widely adopted family of models able to produce photograph-like images by training a generator against a discriminator (Goodfellow et al., 2014). Successive refinements pushed output quality far enough that synthesised human faces hold up under close inspection (Karras et al., 2020). Diffusion models then changed the situation again: instead of an adversarial game, they learn to reverse a gradual noising process (Ho et al., 2020), and the latent-space formulation made text-conditioned generation cheap enough to run on ordinary hardware and to release publicly (Rombach et al., 2022). Producing a convincing synthetic photograph is therefore no longer a specialist skill — it takes a text prompt and a few seconds. Images made this way already circulate on public social platforms in enough volume to be collected into research datasets (Chen and Zou, 2023).

The problems that follow are practical rather than hypothetical. Synthetic imagery can be used to support false claims in reporting, to fabricate apparent evidence of events that never took place, and — because generated faces are convincing enough to have attracted a detection literature of their own (Wang et al., 2023a) — to impersonate people. All of this sits inside the older field of media forensics, which until recently was concerned mainly with edited rather than wholly synthesised images (Verdoliva, 2020). A less obvious consequence is that large image corpora scraped from the web can no longer be assumed to contain only camera output, since the platforms that host synthetic images are the same ones such corpora are drawn from (Chen and Zou, 2023). Recent

surveys treat detection of AI-generated media as a distinct problem for these reasons (Lin et al., 2024).

Manual inspection does not scale to this. The volume of images uploaded to a large platform in a single day is far past what human reviewers could examine, and the evidence a detector actually relies on is often not visible to a viewer at all: much of it consists of statistical traces left by the generator's upsampling operations, which appear in the frequency spectrum rather than in the picture itself (Zhang et al., 2019; Durall et al., 2020). Automated detection is therefore the only workable option. It is also an open research problem rather than a solved one — on a benchmark deliberately built from realistic in-the-wild images, detectors scoring 92.77% mean accuracy on a standard benchmark fall to 65.77% (Yan et al., 2025a).

The central difficulty is generalisation across generators. A detector trained on one generator family usually performs well on that family and considerably worse on others. Early work was encouraging on this point: a ResNet-50 trained only on ProGAN images reached 90.8 mAP averaged over eleven unseen GAN test sets, though the authors qualified the result as holding "for now" (Wang et al., 2020). The arrival of diffusion models largely ended that transfer. GAN-trained detectors lose 15.2% AUROC on average when tested on diffusion images (Ricker et al., 2024a), and accuracy on uncompressed diffusion output has been reported at roughly 60-76%, falling to about 50-61% once images are compressed and resized in the way a social platform would do (Corvi et al., 2023a). The same pattern repeats within the diffusion family: a ResNet-50 reaching 99.9% accuracy on the generator it was trained on scores 54.9% on a different one (Zhu et al., 2023). One reason is that many detection cues are

properties of a particular generator design rather than of synthetic images in general — a widely used spectral cue can be removed by changing the generator's final upsampling layer, taking detection from 84.8-99.8% down to 0-0.3% (Chandrasegaran et al., 2021).

This review surveys published work on detecting AI-generated still images. Its scope is deliberately narrow: single images only, with video and audio deepfake detection excluded; work published between 2019 and 2026; and studies that report quantitative results, so that performance claims can be placed side by side. It differs from existing reviews in two respects. Earlier overviews are either broader in medium, covering audio and video alongside images (Lin et al., 2024), narrower in subject, addressing GAN-generated faces specifically (Wang et al., 2023a), or predate diffusion-based generation altogether (Verdoliva, 2020). Second, cross-generator evaluation and explainability are treated here as properties worth recording per study rather than as discussion points: the comparative table states, for every surveyed work, whether the detector was tested on generators absent from its training data, and what limitation the authors themselves acknowledge.

The contributions of this review are:

1. A taxonomy of detection approaches covering handcrafted and frequency-domain forensics, CNN-based transfer learning, transformer and attention-based methods, multimodal and foundation-model-based detectors, and explainability-oriented work.
2. A comparison of the datasets and benchmarks the field trains and tests on, including CIFAKE (Bird and Lotfi, 2024), GenImage (Zhu et al., 2023) and ArtiFact (Rahman et al., 2023).
3. A review of how performance is reported, noting that accuracy, average precision and AUC are used in varying combinations across studies (Wang et al., 2020; Zhu et al., 2023; Cozzolino et al., 2021), which limits direct comparison.
4. A master comparison table with explicit *Cross-Generator Tested?* and *Key Limitation* columns, so that headline figures are read alongside the conditions that produced them.
5. A gap analysis that motivates the planned second phase of this project: a transfer-learning comparison of four ImageNet-pretrained CNN backbones against a Vision Transformer benchmark (Dosovitskiy et al., 2021) on CIFAKE (Bird and Lotfi, 2024), with Grad-CAM (Selvaraju et al., 2017) used for explanation and a GenImage subset (Zhu et al., 2023) used for cross-generator testing. No experiments are conducted or reported in this review.

The remainder of the paper is organised as follows. Section 2 describes how GANs and diffusion models generate images and what artefacts each tends to leave behind. Section 3 presents the taxonomy of detection approaches. Section 4 compares the datasets and benchmarks in common use, and Section 5 covers evaluation metrics and reporting practice. Section 6 presents the master comparison table and the trends drawn from it. Section 7 sets out the open challenges and the gap analysis, and Section 8 concludes with the case for the proposed follow-on study.

2. Background: GANs, Diffusion Models, and Generator Artefacts

A detector is only intelligible once it is clear what it is looking for. This section describes how the two generator families that dominate the literature actually produce an image, and then traces where the detectable traces in their output come from. The detection methods that exploit those traces are surveyed in Section 3; the concern here is the mechanism, not the detector.

2.1. Generative adversarial networks

A generative adversarial network trains two networks in opposition. The generator maps a random vector to an image. The discriminator is shown a mixture of real and generated images and tries to separate them. The two are updated in alternation, so the generator never receives a target image to copy — its only training signal is the discriminator's current opinion of its output (Goodfellow et al., 2014). Nothing in this arrangement requires the generator to match the statistics of real photographs beyond the point where the discriminator stops complaining, which is the observation Section 2.3 builds on.

Structurally, the generator is a stack of layers that enlarges a small feature map step by step until it reaches full resolution. Later architectures kept that overall shape and changed what happens inside it. StyleGAN2, for instance, redesigned the generator's normalisation and layer structure specifically to remove characteristic artefacts that had been visible in the previous version's output, and reported improved quality on high-resolution face synthesis (Karras et al., 2020). Two consequences follow. Synthetic faces became difficult to dismiss by eye, and generator design became partly an exercise in removing whichever traces someone had recently pointed out.

2.2. Diffusion models

Diffusion models replace the adversarial game with a two-stage construction. The forward process takes a real image and repeatedly adds a small amount of Gaussian noise, over many steps, until the image is indistinguishable from noise. This process is fixed and involves no learning. The model is trained to invert it: given a noisy image and the step it belongs to, predict the noise that was added, so that the step can be undone. Generation then starts from pure noise and applies the learned reverse step repeatedly until an image emerges (Ho et al., 2020). Because each step is a small correction rather than a single leap, training avoids the instability of adversarial optimisation.

Running many denoising steps at full resolution is computationally expensive. Latent diffusion addresses this by moving the process into the compressed latent space of a pretrained autoencoder: the diffusion model operates on a small latent representation, and a decoder expands the final latent back into pixels. The same work introduced cross-attention conditioning, which is the mechanism that lets a text prompt steer generation (Rombach et al., 2022). This is the design underneath the publicly released text-to-image systems, and its split into a latent diffusion process followed by a fixed decoder matters for detection, as the next subsection explains.

2.3. Where the artefacts come from

Both families construct an image by repeated upsampling, and this is the single most studied source of detectable traces.

Convolutional generators using up-convolution operations fail to reproduce the spectral distribution of real images, with the mismatch concentrated in the high frequencies (Durall et al., 2020). Viewed in the discrete cosine transform domain, the same defect appears as regular grid-like patterns present across several GAN architectures (Frank et al., 2020). The upsampling module has been isolated as the cause directly: artefacts characteristic of a GAN pipeline can be reproduced by simulating that module alone, without any real generated images (Zhang et al., 2019). None of this is visible in the picture; it is a property of the frequency spectrum.

The important qualification is that these traces follow from particular design choices rather than from synthesis as such. Replacing the final upsampling layer of a generator with bilinear upsampling largely removes the high-frequency spectral signature that detectors had been keying on (Chandrasegaran et al., 2021). In the same vein, a spectrum classifier trained on artefacts produced by one upsampler does not transfer to GauGAN, which uses a different one (Zhang et al., 2019). The trace can also be suppressed deliberately during training once it is known, by regularising the generator's spectrum (Durall et al., 2020). The frequency behaviour of generative models has therefore been studied as a property of the models themselves, not simply as a convenient detection cue (Schwarz et al., 2021).

At a finer grain, the trace is specific to an individual model rather than to a family. Learned fingerprints extracted from generated images are distinctive enough to attribute an image to the particular network that produced it, and differences confined to the training run — such as the initialisation seed — are reported to yield distinguishable fingerprints (Yu et al., 2019). A hand-crafted counterpart exists: averaging denoising residuals over many images from one source, the same construction used to estimate camera sensor noise in classical forensics, produces a fingerprint that identifies the generating model (Marra et al., 2019). What is being measured in both cases is closer to a serial number than to a general property of synthetic images.

Diffusion models change the picture without erasing it. Compared with GANs they leave fewer detectable traces, and their output degrades detection performance more readily under perturbation (Ricker et al., 2024a). They are not trace-free: forensic analysis of diffusion output finds spectral

fingerprints of the same broad kind (Corvi et al., 2023a), and a comparison spanning thirteen models across GAN, VQ-based, and latent-diffusion designs finds artefacts throughout, differing between families rather than being absent from any of them (Corvi et al., 2023b). Latent diffusion adds a source of its own, since the decoder is fixed and shared across every image a given system produces, leaving a signature that training-free detectors can exploit (Section 3.2) (Ricker et al., 2024b). Finally, all of these traces are fragile. They live in the high frequencies, which is precisely what lossy compression discards — under strong JPEG compression the spectra of generated images become essentially indistinguishable from real ones (Dzanic et al., 2020) — and ordinary post-processing can modify or hide them (Corvi et al., 2023b).

This is what makes generalisation, introduced in Section 1, a structural problem rather than an engineering inconvenience. If the evidence a detector learns is a consequence of one generator's upsampling operator, training run, or decoder, then a new generator that changes any of those does not merely present harder examples: it removes the evidence. The approaches surveyed in Section 3 differ largely in how they attempt to escape this dependence, and Section 7 returns to it as the field's central open challenge.

3. Taxonomy of Detection Approaches

Detectors in this field differ less in what they decide than in where the evidence for the decision comes from, and that is the question this section organises around. The surveyed work is grouped into five families (Fig. 1): measurements designed by hand (3.1), representations learned by a convolutional network trained for the task (3.2), transformer backbones trained or fine-tuned for it (3.3), frozen general-purpose vision-language representations reused for it (3.4), and work whose contribution is an explanation of the decision rather than the decision itself (3.5). Earlier surveys have organised the same literature by medium or by forgery type (Wang et al., 2023a; Verdoliva, 2020); grouping by the origin of the signal is preferred here because it connects directly to the generalisation problem set out in Section 2 — how well a detector survives a change of generator depends mainly on how generator-specific its evidence is. The boundaries are not airtight, and the few studies that sit across two families are flagged where they occur.

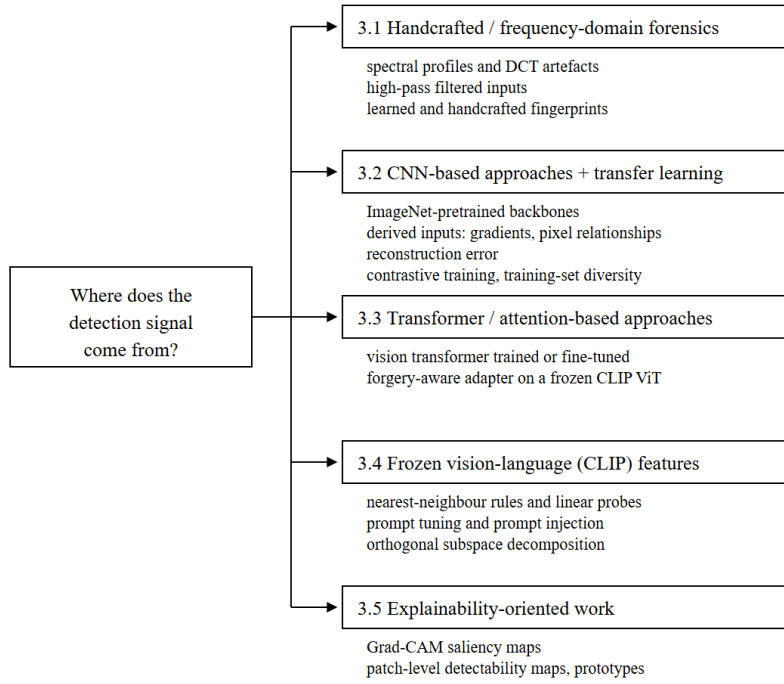


Fig. 1. Taxonomy of detection approaches surveyed in this review, organised by the origin of the detection signal.

3.1. Handcrafted, statistical and frequency-domain forensics

Here a person, not an optimiser, decides what gets measured: the image passes through a fixed transform — Fourier or DCT, a high-pass filter, a denoising residual — and a simple classifier reads the result, continuing classical media forensics practice (Verdoliva, 2020). Section 2 covered why those transforms are mostly spectral; what follows is what detectors built on them reported.

Early results were strong and cheap. Durall et al. (2019) reduced the DFT power spectrum to a one-dimensional profile by azimuthal averaging and classified it with an SVM, logistic regression or k-means, reporting 100% accuracy on their assembled Faces-HQ set from as few as 20 labelled samples. A follow-up extended the feature to CelebA at 100% but reached only 85% on single FaceForensics++ frames, and proposed regularising a generator's spectrum to suppress the trace (Durall et al., 2020). Frank et al. (2020) moved the analysis into the DCT domain, where the upsampling grid is directly visible: ridge regression on DCT coefficients separated real from fake on FFHQ at 100.00%, against 75.78% for the same classifier on raw pixels, though the artefact is shown across the GANs tested rather than by holding one out of training. Dzanic et al. (2020) fitted the high-frequency spectral decay and classified the fitted parameters with a five-nearest-neighbour rule, reaching 99.2% on uncompressed 1024×1024 images.

Two findings qualify this. The same spectral-decay detector falls to 88.8% at 256×256, and under 85% JPEG compression generated spectra become essentially indistinguishable from real ones (Dzanic et al., 2020). Chandrasegaran et al. (2021) retrained generators with bilinear upsampling in the final layer, taking detection built on spectral decay from 84.8–99.8% down to 0–0.3%: the cue follows from a design choice, not from CNN-generated images as such. Zhang et al. (2019) had met a version of this: their classifier, trained on log spectra of artefacts from an AutoGAN simulator rather than

on real fakes, averaged 97.2% accuracy under leave-one-out testing yet failed on GauGAN, which uses a different upsampler. Frequency behaviour is now studied as a property of generative models in its own right (Schwarz et al., 2021).

A later line kept the frequency emphasis but gave the network more work. BiHPF places bilateral high-pass filters ahead of a ResNet-50 for 84.2% mean accuracy on unseen GANs after four-class ProGAN training (Jeong et al., 2022a). FrePGAN inserts a learned frequency-level perturbation before the same backbone, reaching 79.4% average accuracy across models, its authors naming the tradeoff plainly: frequency artefacts help on the training generator and hinder transfer (Jeong et al., 2022b). FreqNet convolves in the FFT domain in a 1.9M-parameter network trained without pretraining, reporting 92.8% mean accuracy over 17 GAN test sets against 83.0% for a CLIP-feature baseline (Tan et al., 2024a; Ojha et al., 2023). NPR swaps the spectrum for a local measurement — relationships between neighbouring pixels left by upsampling — read by a 1.44M-parameter CNN, for 92.2% mean accuracy over 28 GAN, deepfake and diffusion generators (Tan et al., 2024b).

Not every handcrafted feature is spectral. Nataraj et al. (2019) passed pixel co-occurrence matrices to a deep CNN, exceeding 99% accuracy on two GAN datasets by training on one generator family and testing on the other — but only two families were covered. A separate strand measures one specific model, not a family. Marra et al. (2019) averaged denoising residuals over images from one source — the construction used for camera sensor noise — identifying that source at 90.3% accuracy uncompressed and 90.1% after JPEG at quality 95; Yu et al. (2019) learned the fingerprint instead, reaching 99.43% on CelebA in a five-way {real + four GANs} attribution against 86.61% for a PRNU-style baseline. Both are closed-set: they name a source from a known list rather than flag an unfamiliar one. The same style of analysis reaches the diffusion era without a detector attached: Corvi et al. (2023b) compare artefacts across thirteen GAN, VQ and

latent-diffusion models, report no detection metrics, and warn that post-processing may modify and hide them; a dataset-and-detector contribution targets diffusion output directly (Bammey, 2024).

3.2. CNN-based approaches and transfer learning

The alternative is to let a convolutional network find the measurement, usually by transfer learning: an ImageNet-pretrained backbone — ResNet (He et al., 2016), DenseNet (Huang et al., 2017), EfficientNet (Tan and Le, 2019), VGG (Simonyan and Zisserman, 2015) — supplies general features, fine-tuned on real/fake data. Wang et al. (2020) set the family's baseline: a ResNet-50 trained only on ProGAN images with blur and JPEG augmentation averaged 90.8 mAP over eleven unseen-generator test sets, against 90.1 without it. It performs at chance on Photoshop-style manipulations, and its authors qualified the transfer as holding "for now".

Chai et al. (2020) restricted the receptive field so the decision rests on local patches; a truncated Xception classifier trained on ProGAN reached 99.99 AP on StyleGAN, and its per-patch scores double as a map of where the evidence sits, hence its return in Section 3.5.

A cluster of work then tried to make GAN detection universal. Gragnaniello et al. (2021) benchmarked seven detectors, finding AUC above 0.9 on uncompressed unseen GANs but severe degradation after compression and resizing; their own no-downsampling variant exceeded 90% accuracy, yet they judged the field far from reliable tools. Cozzolino et al. (2021) added contrastive training to that design for 95.6% accuracy and 0.997 AUC over seven unseen GAN architectures. Others changed what the network attends to: Gram-Net keys on global texture statistics, above 98% in-domain but around 77.7% trained on StyleGAN and tested on DCGAN, DRAGAN and StarGAN (Liu et al., 2020); a two-branch ResNet-50 fusing global features with attention-selected patches reports 91.73% mean AP on unseen models in a 19-generator set (Ju et al., 2022); and an ensemble of five orthogonally trained EfficientNet-B4 networks (Tan and Le, 2019) reaches 0.9995 AUC on a test set including StyleGAN3, held out of training (Mandelli et al., 2022).

Those numbers came from GANs; diffusion models removed much of the margin. Corvi et al. (2023a) ran GAN-trained detectors against GLIDE, Latent Diffusion, Stable Diffusion, ADM and DALL·E 2 for roughly 60–76% accuracy uncompressed, falling to about 50–61% under compression and resizing. Ricker et al. (2024a) measured the same collapse as an average 15.2% AUROC drop, and its asymmetry: a diffusion-trained detector reached 94.26% average Pd@1%FAR on GANs, a GAN-trained detector 26.34% on diffusion images.

One response keeps the classifier but feeds it a map derived from the image, as NPR does in Section 3.1 (Tan et al., 2024b). LGrad converts an image into gradients of a pretrained CNN before classification, reporting 86.3% mean accuracy over eight test models in its single-class setting, 11.4 points ahead of FrePGAN, trading dependence on training data for dependence on the transformation model (Tan et al., 2023). For diffusion images that map is usually a reconstruction error: DIRE gives a ResNet-50 the residual between an image and its DDIM reconstruction, for 99.9% average accuracy over eight diffusion models (Wang et al., 2023b); SeDID uses per-step denoising error, its network variant averaging 0.9634 AUC across its three test sets (Ma et al., 2023); LaRE² computes that error in latent space to refine

features, up to 11.9% accuracy above the previous best over eight GenImage generators (Luo et al., 2024); and AEROBLADE drops the classifier altogether, scoring an image by how well a latent diffusion model's own autoencoder reconstructs it, for 0.992 mean AP but needing access to that autoencoder (Ricker et al., 2024b). FakeInversion trains a ResNet-50 from scratch on the image plus the noise map and reconstruction from inverting Stable Diffusion, averaging 0.759 AUC-ROC over unseen text-to-image models (Cazenavette et al., 2024).

The remaining studies change the training signal, not the input. DRCT builds hard positives by reconstructing real images through a diffusion model and training contrastively on them, for over 10% accuracy improvement in cross-set tests, less on GAN images (Chen et al., 2024). ZED avoids fake training data entirely, learning a lossless coding model from real images and flagging those it codes unusually cheaply — about 90.0% average AUC across 19 generators, but unreliable on heavily compressed or resized inputs (Cozzolino et al., 2024a). Community Forensics works from the data side, training on 2.7M images from 4,803 generators for 0.987 mAP and 89.2% accuracy on a 21-model out-of-distribution set — error rates the authors still call too high for critical applications (Park and Owens, 2025).

The trend is away from training a convolutional network to find forensic features from scratch and towards a large frozen general-purpose representation — DRCT already uses a CLIP-ViT-L/14 backbone (Chen et al., 2024) and the Community Forensics classifier is a CLIP-pretrained ViT-S (Park and Owens, 2025) — the subject of Sections 3.3 and 3.4.

3.3. Transformer and attention-based approaches

A vision transformer treats an image as a sequence. It is cut into fixed-size patches, each patch is linearly embedded, and self-attention then lets every patch attend to every other from the first layer onward (Dosovitskiy et al., 2021). This is the opposite of the convolutional arrangement, in which a wide view of the image is assembled only gradually by stacking local filters. For detection the difference cuts both ways. The evidence surveyed in Sections 3.1 and 3.2 is mostly local and high-frequency, and several of the stronger convolutional detectors work by protecting exactly that detail — restricting the receptive field to small patches (Chai et al., 2020), or removing the first-layer downsampling that would otherwise discard it (Cozzolino et al., 2021; Gragnaniello et al., 2021). A transformer's patch embedding is itself a stride-16 projection, so it begins by throwing away some of what those methods take care to keep. What it gains is a representation built on relationships across the whole image rather than on filter responses inside a window.

The one surveyed study that isolates the architecture is by Coccomini et al. (2022), who trained a ViT-Base and an EfficientNetV2-M on ForgeryNet subsets and tested both across all fifteen manipulation methods in that dataset. The CNN was the better in-domain classifier, at 81.1% on methods it had seen, but fell to 47.0–57.0% on unseen ones; the ViT stayed at roughly 62.0% in all three of their cross-forgery settings. Neither result is good, and the authors say so — accuracy on novel methods remains low. The pattern is nonetheless the one that recurs elsewhere: the transformer gives up in-domain accuracy and loses less when the forgery changes.

The other primary study does not train a transformer for detection so much as adapt one that already exists. FatFormer keeps a CLIP ViT and attaches a forgery-aware adapter that works over image and frequency representations together, with the model's text side used to align them (Liu et al., 2024a). Trained on the standard four-class ProGAN set, it reports 98% accuracy on unseen GANs and 95% on unseen diffusion models. Because its representation comes from a frozen vision-language model it is surveyed again in Section 3.4; what belongs here is the observation that the useful transformer in this literature is almost never a randomly initialised one.

Studies surveyed elsewhere in this review support that observation. DRCT, covered in Section 3.2 for its contrastive training scheme, is built on a CLIP-ViT-L/14 backbone as well as a ConvNeXt-B one (Chen et al., 2024); Community Forensics, also in Section 3.2, uses a CLIP-pretrained ViT-S as its classifier with a ConvNeXt-S alternative (Park and Owens, 2025); the AIDE detector introduced alongside the Chameleon benchmark in Section 4 pairs DCT scoring and SRM noise filters with an OpenCLIP semantic encoder, giving the transformer the high-level half of a hybrid (Yan et al., 2025a); and the prototype method discussed in Section 3.5 classifies fine-tuned ViT features with an SVM (Aghasanli et al., 2023). A secondary study applies contrastive training with global and local similarities to a large diffusion-image collection released with it (Baraldi et al., 2024).

The caveat is that none of these comparisons is clean. Changing the backbone in this literature usually changes the pretraining data, the training objective and the amount of supervision at the same time, so what the studies establish is that a transformer-based system performed well, not that the transformer is the reason. Only Coccomini et al. (2022) hold the rest of the setup fixed. That is one reason the second phase of this project treats a ViT as a benchmark against CNN backbones under matched training conditions rather than as a proposed improvement.

3.4. Multimodal and foundation-model-based approaches

CLIP is not a detector. It is a pair of image and text encoders trained on image-caption pairs so that matching pairs land close together, giving a visual representation that transfers to tasks it was never trained for (Radford et al., 2021). The work here rests on an empirical finding about that representation: it separates real from generated images although nothing in its training concerned forensics.

Ojha et al. (2023) established the result. They left the CLIP ViT-L/14 image encoder frozen and did no more than read its features, classifying them with a nearest-neighbour rule or a linear probe fitted on the usual ProGAN training set. Against detectors that train their own backbone on the same data this gained 15.07 mAP and 25.90% accuracy on unseen diffusion and autoregressive models. The authors state plainly that why the frozen feature space behaves this way remains an open question; the work that follows takes the representation as given and asks how best to read it.

The answers fall into a few strategies. The lightest is to fit a small classifier on the frozen features: Cozzolino et al. (2024b) take next-to-last-layer CLIP features and fit a linear SVM on as few as ten paired reference images from a single generator, reporting about 90.0% average AUC over eighteen generators, with gains of 6% AUC out of distribution and 13% on impaired images. A second strategy is to read more of the network: RINE uses the intermediate encoder blocks rather

than the final one alone and learns how much each should count, reaching 91.5% average accuracy and 98.8 AP across twenty test sets under four-class training (Koutlis and Papadopoulos, 2024). A third adds trainable modules on top of the frozen model — the FatFormer adapter already described in Section 3.3, at 98% and 95% accuracy on unseen GAN and diffusion images (Liu et al., 2024a), and the prompt tuning that Khan and Dang-Nguyen (2024) compare against linear probing, full fine-tuning and adapters, obtaining 98.06% mAP and 89.45% average accuracy over eighteen unseen generators. C2P-CLIP works on the text side instead, injecting a category-common prompt into the text encoder to carry detection semantics into the image features, for 93.79% accuracy and 98.66% mAP on the nineteen UniversalFakeDetect subsets and 95.8% accuracy on GenImage (Tan et al., 2025). Effort treats the feature space geometrically, applying an SVD to the frozen features and working in orthogonal subspaces of the result, for 99.41% mAP and 95.19% mean accuracy over the same nineteen subsets (Yan et al., 2025b).

The text prompt can also be an input rather than a mechanism. DE-FAKE addresses text-to-image generation, comparing an image-only ResNet18 with a hybrid classifier over CLIP's image and text encoders; trained on Stable Diffusion, the hybrid reaches 0.932, 0.899 and 0.885 accuracy on Latent Diffusion, GLIDE and DALL·E 2, and its authors describe image-only detection on those models as far from their design goal (Sha et al., 2023). Secondary work extends the same ideas to prompt-tuned vision-language models used directly as detectors (Chang et al., 2023), mixtures of low-rank experts (Liu et al., 2024b), language-guided contrastive training (Wu et al., 2026), a captioning reformulation (Keita et al., 2025), and diffusion-image detection with these features (Coccomini et al., 2024).

Two things are missing. The first is any account of what is being measured: the study that opened the line and the few-shot variant both record that the forensic content of CLIP features is not understood (Ojha et al., 2023; Cozzolino et al., 2024b) — the interpretability problem Section 3.5 takes up. The second is robustness. RINE degrades considerably when blur, cropping, compression and noise are combined (Koutlis and Papadopoulos, 2024), and prompt tuning loses accuracy on commercial-tool images under JPEG compression (Khan and Dang-Nguyen, 2024). The protocol is also uniform: several of these detectors are fitted on ProGAN images and reported on diffusion test sets (Ojha et al., 2023; Liu et al., 2024a; Koutlis and Papadopoulos, 2024; Khan and Dang-Nguyen, 2024; Yan et al., 2025b), so the headline generalisation figures describe one particular training-to-test jump rather than the general case.

3.5. Explainability in detection

A detector that outputs a probability says nothing about why. For the forensic uses set out in Section 1 that is a real limitation, and a small body of work therefore asks where in an image the decision comes from. The standard tool is Grad-CAM, which weights the feature maps of a convolutional layer by the gradients of the class score flowing into them and produces a coarse heatmap over the image (Selvaraju et al., 2017); Grad-CAM++ refines that weighting for better localisation, particularly when several instances of a class are present (Chattopadhyay et al., 2018). Both were designed for object recognition, where the highlighted region contains

something a viewer can see. Forensic evidence, as Section 2 established, often is not.

The CIFAKE study makes the mismatch visible. Bird and Lotfi (2024) trained a small custom CNN — two convolutional layers of 32 filters and a 64-unit dense layer, selected from 36 candidate topologies — on their own dataset of 120,000 images, 60,000 real CIFAR-10 photographs and 60,000 generated with Stable Diffusion v1.4, reaching 92.98% accuracy. Applying Grad-CAM to the trained network, they report that the classification is driven by small imperfections in image backgrounds rather than by the subject. The explanation is useful precisely because it is unflattering: what the detector latched onto is a property of one generator at 32-pixel resolution, not a general mark of synthesis.

Chai et al. (2020) reach localisation from the other direction. Their patch classifiers, surveyed in Section 3.2, restrict the receptive field so each decision rests on a small region; collecting the per-patch scores gives a map of which parts of an image are detectable at all. That explains the evidence rather than a particular model's reasoning, and the two are not interchangeable.

Aghasanli et al. (2023) take the interpretable-by-construction route instead of the post-hoc one, classifying fine-tuned ViT features with an SVM and using its support vectors as prototypes a decision can be referred back to. Within-dataset accuracy is high — 99.7% and 96.7% on class-conditional and image-to-image ImageNet sets, 99.4% on FFHQ and 99.5% on Oxford-IIIT Pet with the fine-tuned ViT and an MLP head, and 93.0–97.2% for the SVM on those features — falling to 81.3% and 84.0% in their cross-dataset tests. Those tests are not evidence about unseen generators: both sides were produced with Stable Diffusion, so what changed was the image content, not the generator. The authors also note that they built the datasets themselves for want of open ones.

Whether such explanations are any good has only recently been asked quantitatively. Tsigos et al. (2024) propose an evaluation framework and apply it to an EfficientNetV2 detector on FaceForensics++, comparing Grad-CAM++,

RISE, SHAP, LIME and SOBOL by a common measure: the accuracy drop attributable to the single segment each method ranks most important. LIME ranks best, with that drop running from 0.245 on DeepFakes to 0.579 on NeuralTextures, and the spread across forgery types is itself the finding: Face2Face and FaceSwap remain the hardest to explain. One detector on one dataset is a narrow base, but it establishes that explanation quality is measurable rather than a matter of inspection.

The gap this leaves is between showing where a network looked and stating what it found. A heatmap can expose a shortcut, as in CIFAKE, but it does not say which property of the pixels beneath it is synthetic. The frozen-feature detectors of Section 3.4 sit further from an answer still: their authors record that they do not know what their representation encodes (Ojha et al., 2023; Cozzolino et al., 2024b). CIFAKE is also where this section meets the next: its contribution is a dataset and an explainability analysis together (Bird and Lotfi, 2024), and the datasets and benchmarks the field relies on are the subject of Section 4.

4. Datasets and Benchmarks

4.1. How this review was conducted

The studies surveyed in this paper were collected under a protocol fixed before collection began: six databases (IEEE Xplore, ScienceDirect, SpringerLink, the ACM Digital Library, arXiv and Google Scholar), eight query seeds covering GAN, diffusion and dataset-specific terms, and inclusion limited to still-image work published between 2019 and 2026 that reports quantitative results and appears either in a peer-reviewed venue or as a well-cited preprint. The searches returned 101 records; removing 26 duplicates left 75 unique records for title-and-abstract screening. Of those, 74 were retained — 48 primary studies read in full and entered into an extraction table, 11 secondary studies used selectively, and 15 background references covering generators, backbones and explanation methods — while one preprint was excluded for failing the well-cited criterion. Fig. 2 sets out the flow.

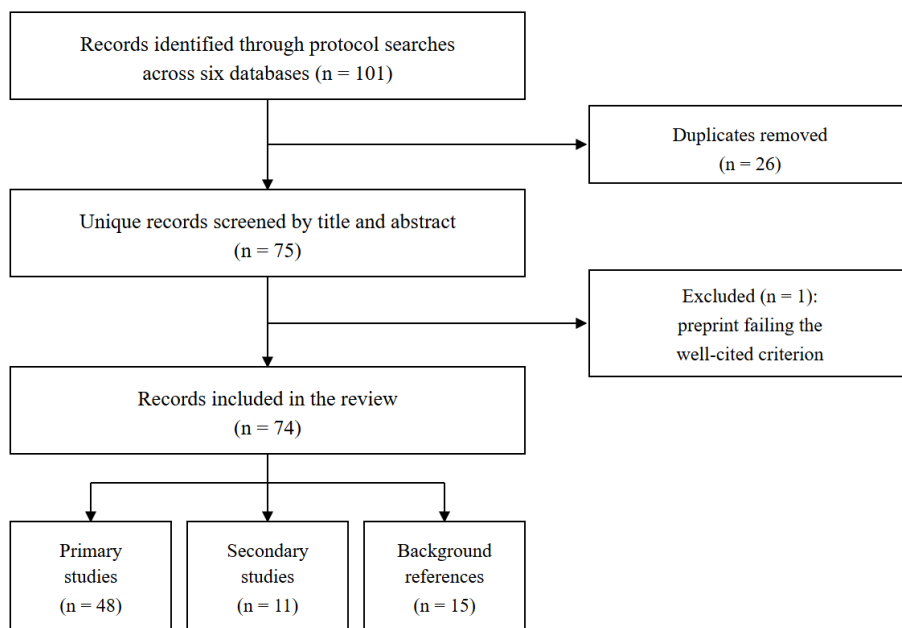


Fig. 2. Flow of records through the review, from database searches to the included set. Counts follow the screening record: 101 identified, 26 duplicates removed, 75 screened by title and abstract, 74 included (48 primary, 11 secondary, 15 background) and 1 excluded.

4.2. From GAN benchmarks to in-the-wild ones

The shape of evaluation in this field was set by ForenSynths, released with the CNN detector of Wang et al. (2020). Its lasting contribution is less the images than the protocol: train on a single generator — 720,000 images drawn from 20 ProGAN categories — and then report performance separately on eleven test sets, one per generator, none of which the detector has seen. That arrangement made cross-generator transfer a headline number rather than an afterthought. Later work reuses the training set directly (Tan et al., 2024a; Tan et al., 2024b; Tan et al., 2023), and several other detectors train on ProGAN images in the same one-, two- or four-class arrangement (Jeong et al., 2022a; Liu et al., 2024a; Koutlis and Papadopoulos, 2024; Khan and Dang-Nguyen, 2024; Yan et al., 2025b). The cost of that convergence is that a large part of the literature shares one training distribution, so the generalisation figures being compared describe a single training-to-test jump rather than the general case. Ojha et al. (2023) extended the same design on the test side, pairing the ProGAN training set with subsets from diffusion and autoregressive models; the nineteen test subsets of the resulting UniversalFakeDetect benchmark are now the standard suite for CLIP-feature detectors (Tan et al., 2025; Yan et al., 2025b).

Diffusion models forced new data. DiffusionForensics, released with the DIRE detector, pairs LSUN-Bedroom and ImageNet reals with eight diffusion models (Wang et al., 2023b). GenImage is the larger effort: over a million

generated images against 1,331,167 ImageNet reals, spanning eight generators from Midjourney and Stable Diffusion to ADM, GLIDE, Wukong, VQDM and BigGAN, and — more importantly — it defines cross-generator classification and degraded-image classification as explicit benchmark tasks (Zhu et al., 2023). ArtiFact is larger still at 2,496,738 images across 25 generation methods, and was built with robustness in mind (Rahman et al., 2023). At the opposite end of the scale sits CIFAKE, 120,000 images built by pairing CIFAR-10 photographs with Stable Diffusion v1.4 output at 32×32 resolution (Bird and Lotfi, 2024); its appeal is that it is small enough to train on modest hardware, which is also its limitation. A further diffusion-era collection, D3, was released alongside a contrastive detection method (Baraldi et al., 2024).

A third generation of datasets gives up on curated generator lists. Community Forensics pushes diversity to its logical end, assembling 2.7 million images from 4,803 generators to test whether breadth of training data alone buys generalisation (Park and Owens, 2025). Chameleon goes the other way, collecting realistic images and reporting detector accuracy on them (Yan et al., 2025a), while TWIGMA takes its images from Twitter, so what it captures is what people actually posted rather than what a researcher chose to generate (Chen and Zou, 2023). Table 1 summarises the datasets discussed here — their scale, the source of their real images, the generators they cover, and the limitation each set of authors states.

Table 1

Datasets and benchmarks on which the surveyed detection literature trains and tests, ordered by year of introduction. "—" marks a property that the review's own extraction records do not state; it is not a claim that the dataset lacks that property.

Dataset	Introduced by	Year	Scale	Real-image source	Generators covered	Notes / stated limitation
ForenSynths (CNNDetection)	Wang et al. (2020)	2020	720K training images (20 ProGAN categories, 18,000 fake + 18,000 real each); 11 generator-specific test sets (Wang et al., 2020; Tan et al., 2023)	LSUN (Wang et al., 2020)	Train: ProGAN. Test: StyleGAN, StyleGAN2, BigGAN, CycleGAN, StarGAN, GauGAN, DeepFakes, CRN, IMLE, SAN, SITD (Wang et al., 2020)	The GAN-era protocol: train on ProGAN, test on unseen generators. Reused as the standard training set by (Tan et al., 2024a; Tan et al., 2024b; Tan et al., 2023); several other detectors train on ProGAN in the same 1-/2-/4-class arrangement (Jeong et al., 2022a; Liu et al., 2024a; Koutlis and Papadopoulos, 2024; Khan and Dang-Nguyen, 2024; Yan et al., 2025b). Authors' own "for now" caveat: later generators may evade it (Wang et al., 2020)
UniversalFakeDetect	Ojha et al. (2023)	2023	720K ProGAN training images; 19 test subsets (Ojha et al., 2023; Tan et al., 2025; Yan et al., 2025b)	— (training set inherited from ForenSynths (Wang et al., 2020))	ProGAN, StyleGAN, BigGAN, CycleGAN, StarGAN, GauGAN, CRN, IMLE, SAN, SITD, DeepFakes, Guided, LDM, Glide, DALL-E (Ojha et al., 2023)	Extends the ForenSynths protocol on the test side with diffusion and autoregressive subsets; its 19 subsets became the standard evaluation suite for CLIP-feature detectors (Tan et al., 2025; Yan et al., 2025b)
DiffusionForensics	Wang et al. (2023b)	2023	—	LSUN-Bedroom, ImageNet (Wang et al., 2023b)	ADM, DDPM, iDDPM, PNDM, LDM, SD-v1, SD-v2, VQ-Diffusion (8 diffusion models) (Wang et al., 2023b)	Released with the DIRE detector; scope limited to diffusion-generated images (Wang et al., 2023b)
GenImage	Zhu et al. (2023)	2023	>1M fake images plus 1,331,167 real (Zhu et al., 2023)	ImageNet (Zhu et al., 2023)	Midjourney, SD v1.4, SD v1.5, ADM, GLIDE, Wukong, VQDM, BigGAN (8) (Zhu et al., 2023)	Defines an explicit cross-generator classification task and a degraded-image task. Authors report sharp degradation cross-generator and on low-resolution, JPEG-compressed and blurred images (Zhu et al., 2023). Later shown to carry JPEG-compression and image-size biases (Grommelt et al., 2024)

Dataset	Introduced by	Year	Scale	Real-image source	Generators covered	Notes / stated limitation
ArtiFact	Rahman et al. (2023)	2023	2,496,738 images (964,989 real / 1,531,749 fake) (Rahman et al., 2023)	—	25 methods: 13 GANs, 7 diffusion, 5 miscellaneous (Rahman et al., 2023)	Built for robustness and used in the IEEE VIP Cup 2022 tests; authors state that state-of-the-art detectors still struggle on generators unseen during training (Rahman et al., 2023)
TWIGMA	Chen and Zou (2023)	2023	—	Not applicable — a dataset of AI-generated images with metadata collected from Twitter (Chen and Zou, 2023)	—	Platform-collected rather than generated for research; screened in as evidence that synthetic images already circulate in the wild (Chen and Zou, 2023)
CIFAKE	Bird and Lotfi (2024)	2024	120,000 images (60,000 real + 60,000 synthetic) (Bird and Lotfi, 2024)	CIFAR-10 (Bird and Lotfi, 2024)	Stable Diffusion v1.4 (latent diffusion) (Bird and Lotfi, 2024)	Single generator at 32×32 resolution; Grad-CAM analysis shows the classification is driven by imperfections in image backgrounds (Bird and Lotfi, 2024)
D3	Baraldi et al. (2024)	2024	—	—	—	Large diffusion-image collection released with a contrastive detection method (ECCV 2024); screened in as a secondary study, so no full-text extraction row exists (Baraldi et al., 2024)
Chameleon	Yan et al. (2025a)	2025	—	—	—	In-the-wild benchmark of realistic images. Detector accuracy falls from 92.77% on AIGCDetectBenchmark and 86.88% on GenImage to 65.77% on Chameleon; the authors state the problem is far from solved (Yan et al., 2025a)
Community Forensics	Park and Owens (2025)	2025	2.7M images from 4,803 generators (Park and Owens, 2025)	—	4,803 generators spanning latent and pixel diffusion, GANs, and commercial models (Park and Owens, 2025)	Built to test whether generator diversity buys generalisation. Authors note the dataset is skewed toward diffusion models and that error rates remain too high for critical applications (Park and Owens, 2025)
WildFake	Hong et al. (2025)	2025	—	—	—	Hierarchical dataset of AI-generated images; screened in as a secondary study for this table only, so no full-text extraction row exists (Hong et al., 2025)

4.3. Known problems with the evaluation data

The most direct problem is that a detection dataset can contain a shortcut. Grommelt et al. (2024) examined GenImage and found that its real and generated images differ systematically in JPEG compression and in image size, so a classifier can score well by reading those properties instead of any forensic trace. Removing the bias costs in-domain accuracy but improves what matters: cross-generator accuracy on GenImage rose by more than eleven points, from 71.68% to 82.74% for a ResNet50 and from 74.09% to 85.83% for a Swin-T. The authors add the appropriate caveat, that other undesirable biases such as style differences and generator fingerprints may remain even after these two are controlled.

Single-generator datasets pose a narrower version of the same problem. CIFAKE contains exactly one generator, and the Grad-CAM analysis published with it reports that the trained network keys on small imperfections in image backgrounds (Bird and Lotfi, 2024). That is a property of Stable Diffusion v1.4 at 32×32 pixels, not a general mark of synthesis, so an accuracy figure obtained on it says little about a different generator. GenImage quantifies the gap on a like-for-like basis: a ResNet-50 reaching 99.9% accuracy when trained and tested on the same generator drops to 54.9% when tested on Midjourney (Zhu et al., 2023). A dataset with one generator cannot expose that drop at all, which is why the Semester 8 plan pairs CIFAKE with a GenImage subset rather than relying on CIFAKE alone.

Staleness follows from the same structure. Every dataset fixes its generator list at publication, and the field's response

has been to keep enlarging the roster — nineteen generators in one recent evaluation (Cozzolino et al., 2024a), eighteen in another (Cozzolino et al., 2024b), 4,803 in Community Forensics (Park and Owens, 2025) — which keeps the benchmarks moving but never quite closes the gap to whatever was released last month. Comparability suffers in a second way when authors build their own data: several surveyed studies evaluate on sets assembled for the paper (Durall et al., 2020; Durall et al., 2019; Ju et al., 2022; Aghasanli et al., 2023), and Aghasanli et al. (2023) state the reason plainly, that open datasets for their setting did not exist.

The sharpest evidence that benchmark performance overstates practical performance comes from Chameleon. Detectors evaluated on it fall from 92.77% mean accuracy on a standard benchmark, and 86.88% on GenImage, to 65.77% on Chameleon's realistic images; the authors conclude that the problem is far from solved (Yan et al., 2025a). Twenty-seven points separate the number a detector reports from the number it earns on images of the kind it would actually meet. That gap is a property of the data, not of any one detector, and it is the reason Section 5 turns next to how performance is measured and reported.

5. Evaluation Metrics and Reporting Practice

5.1. The metrics

At the point of evaluation every detector surveyed here is a binary classifier, and the convention in this literature is to treat the AI-generated image as the positive class. Each decision then falls into one of four counts: a true positive (TP) is a

generated image called generated, a false positive (FP) a real image called generated, a true negative (TN) a real image called real, and a false negative (FN) a generated image called real. Every metric below is a different summary of those four numbers. Fig. 3 summarises the confusion matrix and the quantities derived from it.

Accuracy is the proportion of correct decisions:

$$Accuracy = (TP + TN) / (TP + FP + FN + TN) \quad (1)$$

It is the most frequently reported figure in this field, which is defensible only because the datasets involved are usually balanced by construction — CIFAKE, for instance, pairs 60,000 real with 60,000 generated images (Bird and Lotfi, 2024). On a skewed test set the same number would mostly describe the class prior.

Precision and recall split the errors by type. Precision asks what fraction of the images flagged as generated really were; recall, also called the true positive rate, asks what fraction of the generated images were caught:

$$Precision = TP / (TP + FP) \quad (2)$$

$$Recall = TPR = TP / (TP + FN) \quad (3)$$

The two trade against each other, and the F1 score reports their harmonic mean, which is low unless both are high:

$$F1 = 2 \cdot (Precision \cdot Recall) / (Precision + Recall) \quad (4)$$

All four depend on where the decision threshold is placed. The remaining metrics remove that dependence by sweeping the threshold across its whole range. Writing the false positive rate as

$$FPR = FP / (FP + TN) \quad (5)$$

the receiver operating characteristic (ROC) curve plots TPR against FPR over that sweep, and the area under it is

$$AUC = \int TPR(FPR) d(FPR), \text{ integrated from } FPR = 0 \text{ to } 1 \quad (6)$$

An AUC of 0.5 corresponds to chance and 1.0 to perfect separation. Average precision (AP) applies the same idea to the precision-recall curve, summing the precision at each recall level weighted by the increase in recall since the previous one:

$$AP = \sum_k (Recall_k - Recall_{\{k-1\}}) \cdot Precision_k \quad (7)$$

and where a detector is evaluated on several test sets — normally one per generator — the mean over them is reported as mean average precision:

$$mAP = (1/N) \sum_i AP_i \quad (8)$$

Confusion matrix (positive class: AI-generated image)

	Predicted: generated	Predicted: real
Actual: generated	TP true positive	FN false negative
Actual: real	FP false positive	TN true negative

Metrics derived from the four counts

$$Accuracy = (TP + TN) / (TP + FP + FN + TN)$$

$$Precision = TP / (TP + FP)$$

$$Recall = TP / (TP + FN)$$

$$F1 = 2 \cdot (Precision \cdot Recall) / (Precision + Recall)$$

AUC (area under the ROC curve) and AP (average precision) summarise performance over a sweep of decision thresholds, not at one fixed point.

Fig. 3. Confusion matrix for real-versus-generated classification and the metrics derived from it. The positive class is the AI-generated image. AUC and AP summarise performance over a sweep of decision thresholds rather than at a single operating point.

5.2. How the surveyed literature reports

Several specialised conventions recur. The most common is mAP computed over per-generator test sets, introduced with the ForenSynths protocol and reported there as 90.8 mAP averaged over eleven unseen generators (Wang et al., 2020). A second is the accuracy-and-AUC pair, quoting a threshold-dependent and a threshold-free figure together, as in the 95.6% accuracy and 0.997 AUC reported over seven unseen GAN architectures (Cozzolino et al., 2021). Patch-level and

single-transfer results are often given as AP alone, for example 99.99 AP training on ProGAN and testing on StyleGAN (Chai et al., 2020). The most operationally explicit convention is the probability of detection at a fixed false alarm rate: Ricker et al. (2024a) report Pd@1%FAR, giving 94.26% average detection for a diffusion-trained detector on GAN images against 26.34% for the reverse direction.

This variety is the problem. Different studies report different metric combinations over different test suites: mAP over eleven GAN sets (Wang et al., 2020), accuracy and AP

over eight diffusion models (Wang et al., 2023b), AUC-ROC averaged over unseen text-to-image models (Cazenavette et al., 2024), AUC over nineteen generators (Cozzolino et al., 2024a), and mAP together with accuracy over a twenty-one-model evaluation set (Park and Owens, 2025). Because the suites differ, two headline numbers are rarely measuring the same thing, and a reader cannot recover one metric from another after the fact.

Thresholds compound this. Accuracy, precision, recall and F1 all assume an operating point that is usually left at the default and almost never calibrated across datasets, so an accuracy figure carries an unstated choice with it; AUC and AP avoid the choice but say nothing about performance at the point a deployed system would actually use. Pd@1%FAR (Ricker et al., 2024a) is the exception that shows what is missing elsewhere.

The averages hide as much as they show. A mean over generators can be carried by easy cases while concealing a total failure: the spectrum classifier of Zhang et al. (2019) averages 97.2% accuracy under leave-one-out testing yet fails

on GauGAN, and on GenImage a ResNet-50 at 99.9% on its training generator scores 54.9% on Midjourney (Zhu et al., 2023). The worst-case generator, not the mean, is what a forensic user meets. For this reason the comparative table in Section 6 records the metric, the test suite and the cross-generator condition alongside every figure, rather than the figure alone.

6. Comparative Analysis

Sections 3 to 5 treated detection approaches, datasets and metrics one at a time. This section places the surveyed work side by side. Table 2 lists all 48 primary studies in order of publication, recording for each the approach, the model, the data, the generators covered, the headline figure, whether the detector was evaluated on generators absent from its training data, and the limitation its own authors state. The last two columns are what this review adds to the conventional comparison, and every count below was obtained by tallying a column of the table, so each can be rechecked against it.

Table 2

Comparative summary of the 48 primary studies surveyed in this review, ordered by year of publication and then by screening ID. Each row compresses one record of the review's full-text extraction table; the *Cross-Generator Tested?* column records whether the study evaluated its detector on generators absent from its training data, and *Key Limitation* records the limitation the study's own authors state rather than one attributed by this review.

Study (Author, Year)	Detection Approach	Model / Architecture	Dataset Used	Generator(s) Covered	Reported Accuracy / AUC	Cross-Generator Tested?	Key Limitation
Yu et al., 2019	Learned GAN fingerprints for detection and closed-set source attribution	Deep CNN attribution classifier (fingerprint = final 1x1x512 feature/weight tensors); AutoEncoder visualisation variant	CelebA and LSUN bedroom (20,000 reals each; 100K train / 10K test per source)	ProGAN, SNGAN, CramerGAN, MMDGAN (all 128x128x3)	99.43% (CelebA) / 98.58% (LSUN) on 5-class {real + 4 GANs} attribution, vs PRNU baseline 86.61 / 67.84	No — closed-set attribution, known GANs	Needs source-labelled samples from every candidate GAN; evaluated only at 128x128
Marra et al., 2019	PRNU-style handcrafted fingerprint: averaged denoising residuals, correlation-based source identification	Denosing-filter residual averaging (no learned classifier)	1,000 RGB 256x256 images per source; RAISE camera images; CycleGAN / ProGAN / StarGAN category sets	CycleGAN (9 configs), ProGAN (6 configs), StarGAN (5 configs)	Source-attribution accuracy 90.3% uncompressed, 90.1% after JPEG QF=95	No — source ID among known GANs	Weaker on StarGAN male/smiling networks and on real cameras (low-energy PRNU patterns)
Zhang et al., 2019	Frequency-spectrum classifier targeting GAN upsampling artefacts; AutoGAN simulator generates artefacts without real fakes	ResNet-34 (ImageNet-pretrained) on log frequency spectrum (phase discarded)	CycleGAN 14-category image sets + 4,000 MSCOCO images for AutoGAN training	CycleGAN (primary), StarGAN, GauGAN tested	97.2% avg accuracy (leave-one-out) for the AutoGAN-trained spectrum classifier that never saw real fakes; 99.8 CycleGAN-trained	Yes — unseen GANs via simulated artefacts	All classifiers fail on GauGAN (different upsampler); sensitive to upsampler mismatch
Nataraj et al., 2019	Pixel-domain co-occurrence matrices fed to a deep CNN	Deep CNN on 3-channel co-occurrence matrices	Two GAN datasets, 56,000+ images (unpaired translation; facial attributes)	CycleGAN, StarGAN	>99% classification accuracy on both datasets	Yes — trained one GAN, tested the other	Only two GAN families evaluated; no explicit limitation stated in abstract
Durall et al., 2019	Frequency-domain forensics: azimuthally averaged 1D power spectrum + classical classifier	SVM (RBF), logistic regression, K-means (unsupervised)	Faces-HQ (40K, self-assembled), CelebA, FaceForensics++	Self-trained DCGAN; fakes from thispersondoesnotexist.com; FF++ manipulations	100% accuracy on Faces-HQ with as few as 20 labelled samples; 91% accuracy on FF++ videos	No — per-dataset evaluation only	Low-resolution content harder to identify since the available frequency spectrum is much smaller
Wang et al., 2020	Binary real/fake CNN classifier with careful pre-/post-processing and Blur+JPEG augmentation	ResNet-50 (ImageNet-pretrained)	ForenSynths (720K ProGAN train images; 11-generator test sets; LSUN reals)	ProGAN (train); tested on StyleGAN, StyleGAN2, BigGAN, CycleGAN, StarGAN, GauGAN, DeepFakes, CRN, IMLE, SAN, SITD	mAP 90.8 averaged over 11 unseen-generator test sets with Blur+JPEG(0.5) augmentation (90.1 without)	Yes — ProGAN-trained, 11 unseen test sets	Performs at chance on traditional (Photoshop-style) manipulations; authors' "for now" caveat
Chai et al., 2020	Patch-based classification with limited receptive fields; heatmaps of which patches are detectable	Truncated Xception and ResNet patch classifiers (Xception blocks best)	CelebA-HQ (24,183 train), FFHQ (60,000 train), FaceForensics++	ProGAN, StyleGAN, StyleGAN2, Glow, GMM face generator; FF++ manipulations	AP 99.99 training on ProGAN and testing on StyleGAN; AP ~100 across unseen resolutions	Yes — ProGAN-trained; StyleGAN/Glow/GMM tested	"No method is completely bulletproof" — better generators, out-of-distribution images or adversarial attacks can defeat it
Frank et al., 2020	DCT frequency-domain analysis of grid-like upsampling artefacts with simple classifiers	Ridge regression and a shallow 4-conv-layer CNN (~170K params) on DCT spectra	FFHQ, CelebA, LSUN bedrooms	StyleGAN, BigGAN, ProGAN, SN-DCGAN, CramerGAN, MMDGAN	Ridge-Regression-DCT 100.00% real-vs-fake on FFHQ (vs 75.78% on pixels); CNN-DCT source ID 99.64% LSUN / 99.07% CelebA	No — no train-on-one/test-on-unseen-GAN experiment	Classifier can still be evaded by specifically crafted adversarial perturbations

Study (Author, Year)	Detection Approach	Model / Architecture	Dataset Used	Generator(s) Covered	Reported Accuracy / AUC	Cross-Generator Tested?	Key Limitation
Durall et al., 2020	Azimuthal integration of the DFT power spectrum into a 1D feature + classical classifiers; proposes spectral regularisation	SVM, logistic regression, K-means (unsupervised) on the spectral feature	Faces-HQ (40K, self-built), CelebA (202,599), FaceForensics++	DCGAN, DRAGAN, LSGAN, WGAN-GP (CelebA); FaceForensics++ deepfakes	100% on Faces-HQ (SVM, 1000 samples) and 100% on CelebA (2000 samples); 85% on FF++ single frames	No — per-dataset supervised/unsupervised tests only	Detection assumes up-convolution generators; other generation techniques may not show these signatures
Liu et al., 2020	Global texture (Gram) statistics for fake face detection	Gram-Net (CNN leveraging global texture representations)	CelebA-HQ, FFHQ, CelebA (real); GAN face sets; ImageNet / BigGAN (natural)	StyleGAN, PGGAN, DCGAN, DRAGAN, StarGAN, BigGAN	~77.7% accuracy cross-GAN (train StyleGAN, test DCGAN / DRAGAN / StarGAN); >98% in-domain	Yes — unseen GAN models evaluated	Analysis covers only texture contrast; real/fake differences go beyond it
Dzanic et al., 2020	High-frequency Fourier spectral-decay analysis (fitted decay parameters)	KNN (k=5) on the fitted decay parameters	FFHQ + Cat Head Detection dataset (~5,000 images, 10/90 train/test)	StyleGAN, StyleGAN2, PGGAN, VQ-VAE2, ALAE	99.2% accuracy on uncompressed 1024x1024 images, mixed-generator training	No — five generators, none held out	At 85% JPEG compression generated spectra become essentially indistinguishable from real; 88.8% at 256x256
Gragnaniello et al., 2021	Critical benchmark of state-of-the-art GAN-image detectors under compression, resizing and unseen generators	Seven detectors: Xception, SRNet, Spec, M-Gb, Co-Net, Wang2020 ResNet50, PatchForensics + proposed no-down variant	362K LSUN reals + 362K ProGAN fakes (train); test reals from ImageNet, COCO, RAISE (~39K synthetic test)	ProGAN (train); tested StyleGAN, StyleGAN2, BigGAN, CycleGAN, StarGAN, RelGAN, GauGAN	AUC above 0.9 for several methods on uncompressed unseen GANs; severe drops after compression/resizing; best no-down variant above 90% accuracy	Yes — ProGAN-trained, 7 unseen GANs	Authors state "we are still very far from having reliable tools for GAN image detection"; robustness experiments preliminary
Cozzolino et al., 2021	Contrastive learning with limited sub-sampling architecture	ResNet50 (ImageNet-pretrained, no first-layer subsampling) + 2 projection layers	LSUN (362K real) + 362K ProGAN images (train); ImageNet, COCO, RAISE (test)	ProGAN (train); StyleGAN, StyleGAN2, BigGAN, CycleGAN, StarGAN, RelGAN, GauGAN (unseen test)	95.6% accuracy / 0.997 AUC averaged over unseen GAN architectures	Yes — seven unseen GAN architectures	Robustness to adversarial attacks left to future work
Chandrasegaran et al., 2021	Counter-study of high-frequency spectral-decay features, showing the artefact is removable via upsampling choice	KNN (k=5) on 3 spectral features (also SVM, MLP); GANs retrained with modified last-layer upsampling	CelebA (128), LSUN Bedrooms (128), CelebA-HQ (512)	DCGAN, LSGAN, WGAN-GP, StarGAN (+CRN, IMLE supplementary)	Detection rate collapses to 0–0.3% with bilinear upsampling vs 84.8–99.8% baseline	No — classifiers trained per-GAN	Its own finding: high-frequency spectral decay is not intrinsic to CNN-generated images, so detectors built on it are unreliable
Jeong et al., 2022b	Frequency-level perturbation maps (FrePGAN) to suppress frequency artefacts	Perturbation generator + ResNet-50 deepfake classifier	ProGAN-generated train set; test images from FFHQ, LSUN, ImageNet, CelebA, COCO, Deepfake sets	ProGAN, StyleGAN, StyleGAN2, BigGAN, CycleGAN, StarGAN, GauGAN	79.4% average accuracy / 87.2% average precision cross-model (4-class training)	Yes — known and unseen GAN models	Tradeoff: frequency artefacts aid known-GAN detection but reduce cross-GAN generalisation
Mandelli et al., 2022	Ensemble of orthogonally trained CNNs favouring real-image scores	Ensemble of five EfficientNet-B4 CNNs	FFHQ, Metfaces, AFHQ2 (~116K real) + ~200K synthetic (train); StyleGAN3 test pairs	StyleGAN2, StarGAN-v2, Taming Transformers, FaceVid2Vid, score-based models; StyleGAN3 (unseen)	0.9995 AUC on the global test set including unseen StyleGAN3	Yes — StyleGAN3 excluded from training	Orthogonality among the CNNs is only empirically verified; no theoretical grounding
Ju et al., 2022	Fusion of global spatial and local patch features via attention	Two-branch ResNet-50 (ImageNet-pretrained) + patch selection module + multi-head attention	Self-built diverse test set: 128,424 images from 19 generation models	19 models incl. ProGAN, StyleGAN2/3, BigGAN, WGAN, VAEGAN, GauGAN, StarGAN, CycleGAN, Pix2Pix, FF++, Celeb-DF	91.73% mean AP on unseen models (96.91% global AP mixed)	Yes — unseen-model evaluation is central	Resizing high-resolution images makes artefacts harder to detect
Jeong et al., 2022a	Bilateral high-pass filters (frequency-level + pixel-level) amplifying frequency artefacts	ResNet-50 classifier with a BiHPF front-end	LSUN, FFHQ (real); ProGAN images for training	ProGAN, StyleGAN, StyleGAN2, BigGAN, CycleGAN, StarGAN	84.2% mean accuracy / 81.8% mean AP on unseen GAN models (4-class ProGAN training)	Yes — ProGAN-trained, 5 unseen GANs	No limitations stated; evaluation covers GAN generators only
Coccomini et al., 2022	Cross-forgery comparison of a ViT against a CNN classifier	ViT-Base vs EfficientNetV2-M	ForgeryNet (15 manipulation methods)	GAN / reenactment / swap methods incl. FaceShifter, StyleGAN2, Talking Head Video, FS-GAN, Deepfakes, ATVG-net	Accuracy on unseen methods: EfficientNetV2-M 47.0–57.0% (81.1% seen) vs ViT-Base ~62.0% on all three settings	Yes — trained on subsets, tested 15 methods	Accuracies remain rather low on novel unseen methods
Tan et al., 2023	Learning on Gradients (LGrad): gradients from a pretrained CNN as a generalised artefact representation	Pretrained-CNN transformation model converting images to gradients + classifier	ForenSynths / CNNDetection training set: 20 ProGAN categories, 18,000 fake + 18,000 real each; 1-, 2- and 4-class settings	ProGAN, StyleGAN, StyleGAN2, BigGAN, CycleGAN, StarGAN, GauGAN, Deepfake (8 test models)	86.3% mean ACC / 92.7% mean AP over the 8 test models (1-class setting); +11.4% ACC and +13.4% AP over FrePGAN	Yes — cross-model, cross-data, unseen domains	Turns a data-dependent problem into a transformation-model-dependent one (authors' own framing)
Corvi et al., 2023a	Forensic-trace (spectral fingerprint) analysis of diffusion models + evaluation of existing detectors	Spec, PatchForensics, Wang2020 (ResNet-50 with augmentation), Grag2021 (no first-layer downsampling)	Real: COCO, ImageNet, UCID; COCO prompts used for generation	GLIDE, Latent Diffusion, Stable Diffusion, ADM, DALL-E 2 (+ProGAN, StyleGAN2/3, BigGAN, EG3D references)	Accuracy ~60–76% (AUC ~75–93%) on uncompressed diffusion images; falls to ~50–61% accuracy in the compressed/resized social-network scenario	Yes — GAN-trained detectors tested on diffusion	Generalisation remains the main hurdle; GAN-trained detectors work poorly on diffusion images, worse under compression/resizing
Wang et al., 2023b	Diffusion reconstruction error (DIRE): residual between an image and its DDIM reconstruction	ResNet-50 classifier on DIRE maps; ADM (LSUN-Bedroom) reconstructor, DDIM 20 steps	DiffusionForensics benchmark (LSUN-Bedroom, ImageNet sources)	8 diffusion models: ADM, DDPM, iDDPM, PNDM, LDM, SD-v1, SD-v2, VQ-Diffusion	99.9% average accuracy / 100 AP across 8 diffusion models including unseen ones (ADM-trained)	Yes — generalises to unseen diffusion models	No limitations stated; scope limited to diffusion-generated images

Study (Author, Year)	Detection Approach	Model / Architecture	Dataset Used	Generator(s) Covered	Reported Accuracy / AUC	Cross-Generator Tested?	Key Limitation
Sha et al., 2023	Text-to-image fake detection and attribution via image-only and hybrid (image+prompt) classifiers	ResNet18 (image-only); CLIP image + text encoders with a 2-layer MLP (hybrid)	MSCOCO, Flickr30k (real) + fakes from four text-to-image models	Stable Diffusion, Latent Diffusion, GLIDE, DALL-E 2	Hybrid detector trained on SD: accuracy 0.932 / 0.899 / 0.885 on Latent Diffusion / GLIDE / DALL-E 2 (MSCOCO)	Yes — SD-trained, three unseen T2I models	Image-only detection "far from the design goal" on other models such as GLIDE and DALL-E 2
Ma et al., 2023	Stepwise error between the deterministic reverse and denoising diffusion steps (SeDID)	SeDID_Stat (statistical) and SeDID_NNs (neural network) variants	CIFAR-10, TinyImageNet, CelebA (all 32x32)	DDPM only	SeDID_NNs: average AUC 0.9634 / ACC 0.9399 across the three datasets	No — DDPM only, no other families	Dataset-dependent optimal timestep selection; unusual trends on CelebA need further investigation
Corvi et al., 2023b	Forensic frequency/spectral artefact analysis of synthetic images (analysis paper, no detector proposed)	None — spectral analyses (power spectra, autocorrelation, radial/angular spectra)	Real: ImageNet, FFHQ, LAION; synthetic images from 13 models	GANs (BigGAN, StyleGAN2/3, StyleGAN-T, GALIP), VQ models, DMs (Score-SDE, ADM, GLIDE, DALL-E 2, eDiff-I), latent DMs (LDM, SD, DiT)	None reported — qualitative artefact characterisation, no detection metrics	No — no detector trained	Post-processing "may significantly modify and hide" the identified artefacts
Ojha et al., 2023	Real-vs-fake classification without training the backbone: nearest-neighbour or linear probing in a frozen vision-language feature space	CLIP ViT-L/14 visual encoder (last-layer features) + NN or linear probe	ProGAN real/fake training set (720k images); ForenSynths + diffusion / autoregressive test sets	ProGAN, StyleGAN, BigGAN, CycleGAN, StarGAN, GauGAN, CRN, IMLE, SAN, SITD, DeepFakes, Guided, LDM, Glide, DALL-E	+15.07 mAP / +25.90% accuracy over the previous state of the art on unseen diffusion and autoregressive models	Yes — GAN-trained, tested on diffusion/AR	Why the frozen CLIP feature space separates real from fake "remains an open question"
Zhu et al., 2023	Million-scale benchmark defining cross-generator and degraded-image evaluation tasks	ResNet-50, DeiT-S, Swin-T, CNNSpot, Spec, F3Net, GramNet	GenImage: >1M fake pairs + 1,331,167 ImageNet reals	Midjourney, SD v1.4, SD v1.5, ADM, GLIDE, Wukong, VQDM, BigGAN	ResNet-50 accuracy 99.9% same-generator (SD v1.4) but 54.9% tested on Midjourney	Yes — defines a cross-generator classification task	Detectors degrade sharply cross-generator and on low-resolution, JPEG-compressed and blurred images
Rahman et al., 2023	Large-scale dataset + multi-class classification with filter stride reduction	ConvNeXt-Large (modified stem)	ArtiFact: 2,496,738 images (964,989 real / 1,531,749 fake)	25 methods: 13 GANs, 7 diffusion, 5 miscellaneous	Accuracy 96.04% / 83.00% / 90.60% on IEEE VIP Cup 2022 Tests 1–3	Yes — unseen generators in challenge tests	State-of-the-art detectors still struggle on generators unseen during training
Aghasanli et al., 2023	Interpretable prototype-based detection via ViT features + SVM support vectors	Fine-tuned ViT + SVM (repo also uses xDNN, KNN, Naive Bayes)	Custom datasets built from ImageNet, FFHQ, Oxford-IIIT Pet	Diffusion models (Stable Diffusion; various diffusion generative techniques)	Within-dataset best 99.5% (Oxford-IIIT Pet), 96.7% (image-to-image ImageNet), 99.7% (class-conditional ImageNet), 99.4% (FFHQ) with fine-tuned ViT + MLP head; SVM on fine-tuned features 93.0–97.2%; cross-dataset 81.3% and 84.0%	No — cross-dataset only, no unseen-generator test	Lack of open datasets; evaluated only on self-created custom datasets
Tan et al., 2024b	Neighbouring Pixel Relationships (NPR) capturing up-sampling artefacts	Lightweight CNN (conv layers + ResNet blocks, 1.44M params) on NPR	ForenSynths ProGAN 4-class train (car, cat, chair, horse; 18K fake + 18K real per class)	28 generative models: 16 GANs, 1 deepfake method, 11 diffusion models	92.2% mean accuracy across 28 generators (+11.6% over the prior state of the art)	Yes — 28 unseen GAN and diffusion models	No explicit limitation stated; trained solely on ProGAN 4-class data
Tan et al., 2024a	Frequency-space learning: high-frequency image/feature representations + convolution in the FFT domain	FreqNet — lightweight CNN with residual blocks, no pretraining, 1.9M parameters	ForenSynths (ProGAN training, 1/2/4-class) + GAN self-synthesis test sets	17 GANs incl. ProGAN, StyleGAN, StyleGAN2, BigGAN, CycleGAN, StarGAN, GauGAN, AttGAN, plus a Deepfake set	92.8% mean accuracy over 17 GAN test sets, +9.8% over Ojha (83.0%)	Yes — ProGAN-trained, 17 GAN test sources	No limitations stated in the conclusion; evaluation restricted to GAN sources
Ricker et al., 2024a	Benchmark of GAN detectors on diffusion images + frequency-domain artefact analysis + retraining	Wang2020 (ResNet-50), Gragnaniello2021, Mandelli2022 (EfficientNet-B4 ensemble)	LSUN Bedroom 256 (primary); LSUN Cat/Horse, ImageNet, FFHQ extensions	5 GANs (ProGAN, StyleGAN, ProjectedGAN, Diff-StyleGAN2, Diff-ProjectedGAN) + 5 DMs (DDPM, IDDP, ADM, PNDM, LDM)	GAN detectors drop on average 15.2% AUROC on DM images; a DM-trained detector reaches 94.26% average Pd@1%FAR on GANs, a GAN-trained one only 26.34% on DMs	Yes — GAN-trained vs DM-trained cross-evaluation	DMs leave fewer detectable artefacts, and perturbations degrade DM detectability more strongly
Ricker et al., 2024b	Training-free detection via autoencoder reconstruction error of latent diffusion models	Pretrained LDM autoencoders + reconstruction distance (LPIPS); no trained classifier	Real images from LAION-5B + prompt-matched generated counterparts (Midjourney set from Discord)	Stable Diffusion 1.1/1.5/2.1, Kandinsky 2.1, Midjourney 4/5/5.1	Mean AP 0.992 across state-of-the-art LDMs including SD, Kandinsky and Midjourney	Yes — multiple LDMs incl. proprietary Midjourney	Best results require access to the generating LDM's autoencoder; method scoped to latent diffusion
Luo et al., 2024	Latent reconstruction error (LaRE) feature for diffusion-image detection	LaRE feature + Error-Guided feature REfinement module (EGRE, spatial + channel align-then-refine)	GenImage benchmark	8 image generators (GenImage subsets)	Surpasses the state of the art by up to 11.9% / 12.1% average ACC / AP across 8 generators on GenImage	Yes — across 8 GenImage generators	Residual real/fake feature overlap on unseen generators (Midjourney, ADM, BigGAN); LaRE alone insufficient as sole input
Grommelt et al., 2024	Exposing and removing JPEG-compression and image-size biases in detection datasets	ResNet50 and Swin-T detectors	GenImage	Midjourney, SD v1.4, SD v1.5, Wukong, VQDM, ADM, GLIDE, BigGAN	Bias-corrected training raises cross-generator accuracy by >11 pp on GenImage: ResNet50 71.68% -> 82.74%, Swin-T 74.09% -> 85.83%	Yes — cross-generator across GenImage subsets	Even with constraints, "datasets may contain other undesirable biases" (style differences, generator fingerprints)

Study (Author, Year)	Detection Approach	Model / Architecture	Dataset Used	Generator(s) Covered	Reported Accuracy / AUC	Cross-Generator Tested?	Key Limitation
Chen et al., 2024	Contrastive training on diffusion-reconstructed hard samples (DRCT framework)	ConvNeXt-B and CLIP-ViT-L/14 (UnivFD) backbones	DRCT-2M (proposed, million-scale), GenImage; real images from MSCOCO	16 types of diffusion models (incl. SD v1.4 / v2)	Over 10% accuracy improvement in cross-set tests (trained on SD v1.4, tested on GenImage subsets)	Yes — unseen diffusion models, cross-set tests	Improvement on non-diffusion (GAN) images is less marked; covers only globally generated images, not local regions
Cazenavette et al., 2024	Detection via Stable Diffusion inversion features	ResNet50 trained from scratch on the concatenated image + decoded noise map + reconstruction	Real: LAION, WebLI, reverse-image-search sets; new SynRIS benchmark (15 eval sets)	Trained on SD; evaluated on 15+ models incl. DALL-E 2/3, Midjourney, Imagen, Kandinsky 2/3, PixArt-alpha, SDXL, Playground 2.5, Stable Cascade	Average AUC-ROC 0.759 across the tested generators under the RIS protocol (trained on SD+LAION)	Yes — SD-trained, unseen T2I models tested	May not transfer to non-diffusion text-to-image models (GANs, transformers); the inversion pipeline needs significantly more compute
Cozzolino et al., 2024a	Zero-shot entropy-based forensics: coding cost under a model of real images, trained on real images only	Lossless multi-resolution neural image encoder (ZED)	Open Images (~338k real images) for encoder training; 19-generator synthetic test set	19 generators incl. GauGAN, BigGAN, StyleGAN2, GigaGAN, ADM, GLIDE, SD/SDXL, DiT, DALL-E 1/2/3, Midjourney v5, Firefly	~90.0% average AUC across 19 generators; >3% average accuracy gain over the state of the art	Yes — zero-shot, 19 unseen generators	Unreliable on heavily compressed or resized images; detects fully generated images only, not local manipulations
Cozzolino et al., 2024b	Few-shot CLIP-feature forensics: linear SVM on frozen CLIP features from a handful of reference images	CLIP ViT-L/14 (next-to-last-layer features) + linear SVM	Few paired real/fake reference images (10 to 10k) from a single generator; 18-generator test suite	18 generators: ProGAN, StyleGAN2/3/T, GigaGAN, Score-SDE, ADM, GLIDE, eDiff-I, LDM, SD variants, DiT, DeepFloyd-IF, DALL-E 2/3, Midjourney v5, Firefly	~90.0% average AUC across 18 generators; +6% AUC out of distribution and +13% on impaired data vs the state of the art	Yes — 18 generators, 3 families	Interpretability lacking: which forensic features CLIP exploits is not yet understood
Koutlis and Papadopoulos, 2024	Synthetic-image detection from intermediate CLIP encoder-block representations with learnable importance weighting	RINE: frozen CLIP ViT-L/14 intermediate blocks + a lightweight trainable projection/weighting network	ProGAN training data (1/2/4-class settings); 20 test datasets	ProGAN, StyleGAN2/2, BigGAN, CycleGAN, StarGAN, GauGAN, DeepFake, SITD, SAN, CRN, IMLE, Guided, LDM (3), Glide (3), DALL-E	91.5% average accuracy and 98.8 AP (4-class training) across 20 test sets; +10.6% ACC over the state of the art	Yes — 20 test sets, GANs + diffusion	Combined perturbations (blur + crop + compression + noise) considerably reduce performance
Liu et al., 2024a	Forgery-aware adaptive transfer of CLIP: an adapter over image and frequency domains with language-guided alignment	FatFormer (CLIP-ViT + forgery-aware adapter + text-prompt alignment)	ProGAN 4-class training set; unseen GAN and diffusion test sets	GANs and diffusion models (unseen at test time)	98% accuracy on unseen GANs, 95% on unseen diffusion models	Yes — unseen GANs and diffusion models	Trained on ProGAN data only; authors state room to improve on diffusion models such as Guided, and that GAN vs diffusion artefact relationships remain unresolved
Khan and Dang-Nguyen, 2024	Adapting a vision-language model (CLIP) via prompt tuning for universal detection	CLIP ViT (prompt tuning; compared with linear probe, fine-tuning, adapter)	ProGAN 200k train; 21 test datasets	GANs, diffusion models, commercial tools	98.06% mAP / 89.45% average accuracy across 18 unseen generators (prompt tuning)	Yes — 21 unseen test datasets	Performance drops on commercial-tool images under JPEG compression
Bird and Lotfi, 2024	CNN binary real-vs-fake classification with Grad-CAM explainability	Custom CNN (best of 36 topologies: 2 conv layers x32 filters + dense 64)	CIFAKE: 120,000 images (60k CIFAR-10 real + 60k synthetic)	Stable Diffusion v1.4 (latent diffusion)	92.98% accuracy on the CIFAKE test set	No — single generator (SD v1.4) only	Single generator at 32px resolution; classification driven by background imperfections
Tsigos et al., 2024	Quantitative evaluation framework for XAI methods applied to a deepfake detector	EfficientNetV2 detector; Grad-CAM++, RISE, SHAP, LIME and SOBOLO compared	FaceForensics++	Four face forgery types: DeepFakes, FaceSwap, Face2Face, NeuralTextures	LIME best: top-1 segment accuracy drop 0.245 (DeepFakes) to 0.579 (NeuralTextures)	No — single detector on FF++ only	Face2Face and FaceSwap fakes remain the most challenging for explanation methods
Park and Owens, 2025	Supervised detector trained on a massively diverse multi-generator dataset	ViT-S (CLIP-pretrained) classifier; ConvNeXt-S also tested	Community Forensics (2.7M images, 4,803 generators); evaluated on Wang, Ojha, Synthbuster, GenImage + own 21-model eval set	4,803 generators: latent and pixel diffusion, GANs, commercial models	0.987 mAP and 89.2% accuracy on the out-of-distribution eval set (21 unseen models)	Yes — 21 unseen models, 5 families	Dataset skewed toward diffusion models; error rates still too high for critical applications
Yan et al., 2025a	Hybrid low-level (frequency/noise) and high-level (semantic) mixture-of-experts detection; proposes the in-the-wild Chameleon benchmark	AIDE: DCT scoring module + SRM noise filters + OpenCLIP semantic encoder	AIGCDetectBenchmark, GenImage, Chameleon (proposed)	ProGAN, StyleGAN2/2, BigGAN, CycleGAN, StarGAN, GauGAN, ADM, Glide, SD v1.4/1.5, VQDM, Wukong, Midjourney, DALL-E-2/3	92.77% mean accuracy on AIGCDetectBenchmark, 86.88% on GenImage; drops to 65.77% on Chameleon	Yes — 16+ generators across benchmarks	All tested detectors near-fail on realistic Chameleon images; the problem is "far from being solved"
Tan et al., 2025	Category-common prompt injected into the CLIP text encoder to embed detection semantics into image features	C2P-CLIP (CLIP ViT-L/14 with a category common prompt)	UniversalFakeDetect (ProGAN train, 19 test subsets) and GenImage (SD v1.4 train)	20 generation models (UniversalFakeDetect subsets + GenImage diffusion models)	93.79% accuracy / 98.66% mAP on the 19 UniversalFakeDetect test subsets; 95.8% accuracy on GenImage	Yes — 20 generation models	Word-frequency prompt analysis lacks full-caption analysis, giving incomplete information
Yan et al., 2025b	Orthogonal subspace decomposition (SVD) of frozen foundation-model features (Effort)	CLIP ViT-L/14 (BEiT-v2, SigLIP ablations)	ProGAN train; 19 UniversalFakeDetect test subsets (also FF++ for deepfakes)	ProGAN, CycleGAN, BigGAN, StyleGAN, GauGAN, StarGAN, Guided/Latent Diffusion, GLIDE, DALL-E	99.41% mAP / 95.19% mAcc across 19 test subsets including unseen generators	Yes — 19 unseen-generator subsets	No technical limitations discussed; only misuse risk acknowledged

6.1. What the table counts

Cross-generator evaluation is now common practice but not

universal: 36 of the 48 studies report one and 12 do not. The 12 are early per-dataset spectral work (Durall et al., 2020; Dzanic et al., 2020; Durall et al., 2019), closed-set attribution

among known GANs (Yu et al., 2019; Marra et al., 2019), single-generator or single-detector studies (Bird and Lotfi, 2024; Ma et al., 2023; Tsigos et al., 2024), an analysis paper with no detector (Corvi et al., 2023b), a counter-study with per-generator classifiers (Chandrasegaran et al., 2021), a DCT analysis holding no GAN out (Frank et al., 2020), and a prototype method whose cross-dataset test keeps Stable Diffusion on both sides (Aghasanli et al., 2023). Classifying each study by the families its generator list names, and resolving benchmark names to the generator lists recorded in Table 1, 20 of the 48 cover GAN-era generators only, 8 cover diffusion-family generators only, 19 cover both, and one covers face-manipulation methods alone (Tsigos et al., 2024). Fifteen train on the ProGAN distribution introduced with ForenSynths (Wang et al., 2020; Cozzolino et al., 2021; Jeong et al., 2022a; Jeong et al., 2022b; Tan et al., 2024a; Ojha et al., 2023; Tan et al., 2024b; Chai et al., 2020; Gragnaniello et al., 2021; Tan et al., 2023; Liu et al., 2024a; Koutlis and Papadopoulos, 2024; Khan and Dang-Nguyen, 2024; Tan et al., 2025; Yan et al., 2025b), so close to a third of the surveyed work shares a single training set. Reporting is concentrated in a similar way: roughly 38 studies quote an accuracy figure, 14 an average precision, 9 an AUC or AUROC, and one reports no detection metric at all (Corvi et al., 2023b). Around 23 give an accuracy figure and nothing else, which is the practice Section 5 identified as the least informative. Ten name robustness to degraded or low-resolution input — compression, resizing, blur or other post-processing — in their stated limitation (Ricker et al., 2024a; Corvi et al., 2023a; Zhu et al., 2023; Corvi et al., 2023b; Dzanic et al., 2020; Durall et al., 2019; Ju et al., 2022; Cozzolino et al., 2024a; Koutlis and Papadopoulos, 2024; Khan and Dang-Nguyen, 2024).

6.2. The chronological arc

Read in publication order, the table traces a clear arc. The earliest rows measure one artefact and classify it simply, and the numbers are near perfect: 100% on a self-assembled face set from as few as 20 labelled samples (Durall et al., 2019), 100.00% on FFHQ from DCT coefficients (Frank et al., 2020), 99.2% on uncompressed 1024×1024 images (Dzanic et al., 2020), above 99% from co-occurrence matrices (Nataraj et al., 2019). Those figures describe narrow conditions, and the qualifying rows arrive quickly: the same spectral-decay feature falls to 88.8% at 256×256 and becomes unusable under 85% JPEG compression (Dzanic et al., 2020), and retraining generators with bilinear upsampling takes detection built on it from 84.8–99.8% down to 0–0.3% (Chandrasegaran et al., 2021). The transfer-learning CNN line begins in the same window with a broader claim: a ResNet-50 trained only on ProGAN images averaging 90.8 mAP over eleven unseen generators (Wang et al., 2020), raised to 95.6% accuracy and 0.997 AUC by contrastive training (Cozzolino et al., 2021) and to 0.9995 AUC by an ensemble that held StyleGAN3 out of training (Mandelli et al., 2022). Gragnaniello et al. (2021) evaluated seven such detectors together and judged the field far from reliable tools.

Diffusion models then removed most of that margin, and the table records the collapse twice. GAN-trained detectors tested on GLIDE, Latent Diffusion, Stable Diffusion, ADM and DALL·E 2 fall to roughly 60–76% accuracy uncompressed and about 50–61% once images are compressed and resized (Corvi et al., 2023a). The same loss is measured elsewhere as an average 15.2% AUROC drop and

shown to be asymmetric: a diffusion-trained detector reaches 94.26% average Pd@1%FAR on GAN images, a GAN-trained one 26.34% in the reverse direction (Ricker et al., 2024a). Within the diffusion family the problem repeats, a ResNet-50 at 99.9% on its training generator scoring 54.9% on Midjourney (Zhu et al., 2023). The response visible in the 2023–2024 rows is to stop giving the classifier the image. LGrad classifies gradients taken from a pretrained network, for 86.3% mean accuracy over eight test models (Tan et al., 2023); NPR classifies relationships between neighbouring pixels, for 92.2% over 28 generators (Tan et al., 2024b); DIRE classifies the residual against a DDIM reconstruction, for 99.9% average accuracy over eight diffusion models (Wang et al., 2023b); AEROBLADE removes the classifier altogether and scores an image by autoencoder reconstruction distance, for 0.992 mean AP (Ricker et al., 2024b); LaRE² moves that error into latent space (Luo et al., 2024), and FakeInversion into Stable Diffusion's inversion, at 0.759 average AUC-ROC on unseen text-to-image models (Cazenavette et al., 2024).

The most recent rows converge on a different move: leave a large general-purpose representation untouched and fit something small on it. Ojha et al. (2023) established the result with a nearest-neighbour rule and a linear probe on frozen CLIP features, gaining 15.07 mAP and 25.90% accuracy on unseen diffusion and autoregressive models. Later rows read the same features differently — a few-shot linear SVM at about 90.0% average AUC over 18 generators (Cozzolino et al., 2024b), intermediate encoder blocks at 91.5% accuracy and 98.8 AP over 20 test sets (Koutlis and Papadopoulos, 2024), an adapter at 98% and 95% accuracy on unseen GANs and diffusion models (Liu et al., 2024a), prompt tuning at 98.06% mAP over 18 unseen generators (Khan and Dang-Nguyen, 2024), prompt injection at 93.79% accuracy on 19 test subsets (Tan et al., 2025), and an orthogonal subspace decomposition at 99.41% mAP and 95.19% mean accuracy on the same 19 (Yan et al., 2025b).

6.3. What the strong cross-generator rows have in common

Three properties recur among the rows with the strongest unseen-generator figures. The first is that the representation is not trained on detection data: the foundation-model rows keep the backbone frozen and train only a probe, an adapter or a prompt (Ojha et al., 2023; Liu et al., 2024a; Cozzolino et al., 2024b; Koutlis and Papadopoulos, 2024; Khan and Dang-Nguyen, 2024; Tan et al., 2025; Yan et al., 2025b), and ZED trains only on real images yet reaches about 90.0% average AUC across 19 generators (Cozzolino et al., 2024a). The second is a derived input in place of the pixels — gradients (Tan et al., 2023), neighbouring-pixel relationships (Tan et al., 2024b), reconstruction residuals (Ricker et al., 2024b; Wang et al., 2023b; Luo et al., 2024; Cazenavette et al., 2024). The third is diversity on the training side: Community Forensics trains on 2.7M images from 4,803 generators for 0.987 mAP and 89.2% accuracy on 21 unseen models (Park and Owens, 2025), and removing JPEG-compression and image-size bias from GenImage raises cross-generator accuracy by more than eleven points for both a ResNet50 and a Swin-T (Grommelt et al., 2024). None of the three asks a network to learn a generator-specific artefact from scratch.

6.4. What the headline column hides

Those figures must be read with the two columns beside them. Most describe a single training-to-test jump, usually out of the ProGAN distribution that fifteen rows share, so they measure

one transfer rather than transfer in general. Averages conceal the worst case: a spectrum classifier averaging 97.2% fails outright on GauGAN (Zhang et al., 2019), and LaRE²'s own stated limitation records residual real/fake overlap on Midjourney, ADM and BigGAN (Luo et al., 2024). Benchmark and in-the-wild performance diverge sharply, the same detectors reaching 92.77% and 86.88% on standard benchmarks falling to 65.77% on realistic images (Yan et al., 2025a). The ten degradation-related limitations are borne out by the figures already quoted for compression and resizing (Corvi et al., 2023a), and by a zero-shot detector its authors call unreliable on heavily compressed or resized inputs (Cozzolino et al., 2024a). Sections 4 and 5 examined the dataset and metric reasons; the point here is that the rightmost columns of Table 2 qualify the headline one.

6.5. Where explainability sits

Explanation appears in only four of the 48 rows, and mostly after the fact. Bird and Lotfi (2024) apply Grad-CAM to a small CNN on CIFAKE, finding the decision rests on background imperfections; Chai et al. (2020) obtain a map of which patches are detectable by restricting the receptive field; Aghasanli et al. (2023) build interpretability in, using SVM support vectors over fine-tuned ViT features as prototypes; and Tsigos et al. (2024) compare five explanation methods on one detector, LIME ranking best with a top-1 segment accuracy drop of 0.245 to 0.579. Of those four, only Chai et al. (2020) carries a cross-generator test; the other three are marked No in Table 2. On the evidence of the table, explainability and generalisation are studied separately.

6.6. The combination the table does not contain

Several rows do compare architectures under one protocol: GenImage benchmarks a ResNet-50, a DeiT-S and a Swin-T alongside four purpose-built detectors (Zhu et al., 2023), and Grommelt et al. (2024) run a ResNet50 and a Swin-T through the same bias-correction procedure. Coccomini et al. (2022) come closest to isolating the architecture itself, training a ViT-Base and an EfficientNetV2-M on the same ForgeryNet subsets and testing both across fifteen manipulation methods: the CNN was better in-domain at 81.1% and worse outside it at 47.0–57.0%, against the ViT's roughly 62.0%. None of these compares several ImageNet-pretrained CNN backbones of the kind Section 3.2 described, and none reports an explanation analysis.

The reverse combination is missing too. The single CIFAKE row uses a custom CNN chosen from 36 topologies rather than pretrained backbones, covers one generator, and produces a Grad-CAM analysis but no cross-generator result (Bird and Lotfi, 2024); the prototype-based interpretable row is likewise marked No (Aghasanli et al., 2023); and the one quantitative study of explanation quality covers a single detector on FaceForensics++ (Tsigos et al., 2024). No row in Table 2 therefore combines all four of the following: several ImageNet-pretrained CNN backbones trained under one protocol, a Vision Transformer benchmark trained under the same protocol, an analysis of what the trained models attend to, and a cross-generator test. Each element is well established somewhere in the table; what the surveyed literature does not report is the four together, on a dataset small enough to make the matched comparison tractable. The gap is that combination, not any single element, and Section 7 develops it into the gap analysis.

7. Challenges and Gap Analysis

Sections 3 to 6 described what has been built and how it is measured. This section collects what remains unsolved, as five named sub-problems, and closes with the gap that motivates the planned second phase.

7.1. Cross-generator generalisation

Section 2 argued that this is a structural problem rather than an engineering inconvenience: if the evidence a detector learns is a consequence of one generator's upsampling operator, training run or decoder, then a new generator that changes any of those does not merely present harder examples, it removes the evidence. GAN-era transfer initially looked solid, a ResNet-50 trained on ProGAN alone averaging 90.8 mAP over eleven unseen generators (Wang et al., 2020). Diffusion models ended it: GAN-trained detectors lose 15.2% AUROC on average against diffusion output (Ricker et al., 2024a), with absolute accuracies falling to roughly 60–76% (Corvi et al., 2023a). The problem then reappeared inside the diffusion family, a ResNet-50 at 99.9% accuracy on its own training generator scoring 54.9% on another (Zhu et al., 2023). Finally, the gap separates not only generator families but benchmarks from the images a detector would actually meet: detectors reaching 92.77% mean accuracy on a standard benchmark reach 65.77% on realistic in-the-wild images (Yan et al., 2025a). Section 6.3 identified what the strongest rows share — frozen general-purpose representations, derived inputs, training-set diversity — but none of the three closes the gap.

7.2. Robustness to compression and post-processing

Ten of the 48 studies name robustness to degraded or low-resolution input in their own stated limitation (Section 6.1), the most commonly acknowledged weakness in the table. The cause is the one above: the traces sit in the high frequencies, which is what lossy compression discards. Under 85% JPEG compression the spectra of generated images become essentially indistinguishable from real ones (Dzanic et al., 2020), and ordinary post-processing can modify or hide the artefacts (Corvi et al., 2023b). The effect is measured directly — GAN-trained detectors on diffusion output fall from roughly 60–76% accuracy to about 50–61% once images are compressed and resized (Corvi et al., 2023a), a zero-shot detector at about 90.0% average AUC is called unreliable on heavily compressed or resized inputs by its own authors (Cozzolino et al., 2024a), and a CLIP-feature detector degrades under combined blur, cropping, compression and noise (Koutlis and Papadopoulos, 2024). Compression is not an attack; it is the default behaviour of every platform through which a suspect image would arrive.

7.3. Dataset bias and staleness

A detection dataset can be won without detecting anything. Grommelt et al. (2024) found that real and generated images in GenImage differ systematically in JPEG compression and image size, so a classifier can score well by reading those properties rather than any forensic trace; removing both biases raises cross-generator accuracy by over eleven points for a ResNet50 and a Swin-T. Single-generator datasets pose a narrower version of the same problem: CIFAKE contains one generator at 32×32 resolution, and the Grad-CAM analysis published with it reports that the trained network keys on small imperfections in image backgrounds — a property of that generator, not a general mark of synthesis (Bird and Lotfi,

2024). Such a dataset cannot expose a cross-generator drop at all. Staleness follows from the same structure: every dataset fixes its generator list at publication, and the field's response has been to enlarge the roster rather than escape the format, which is what the benchmark-versus-wild gap (Yan et al., 2025a) eventually costs.

7.4. Interpretability

Explanation appears in only four of the 48 rows (Section 6.5), and what those four mostly provide is post-hoc saliency: a map of where a network looked, not a statement of what it found. The distinction matters more here than in object recognition, because the evidence is frequently invisible in the picture. Grad-CAM on the CIFAKE network exposes a shortcut without naming which property of those pixels is synthetic (Bird and Lotfi, 2024); prototype-based classification is interpretable by construction but was evaluated with the same generator on both sides of its cross-dataset test (Aghasanli et al., 2023); and the one quantitative comparison of explanation methods covers a single detector on a single dataset (Tsigos et al., 2024). The frozen-feature detectors sit furthest from an answer: their authors record that why a CLIP representation separates real from generated remains open (Ojha et al., 2023; Cozzolino et al., 2024b). Exposing a shortcut is useful, but it is not the same as stating what a detector measures.

7.5. Reproducibility and comparability

A smaller problem compounds the others. Section 5 recorded that studies report different metric combinations over different test suites, so two headline figures rarely measure the same thing; around 23 of the 48 quote an accuracy figure and nothing else (Section 6.1). Cross-study comparison is therefore weaker evidence than it appears — an argument for evaluating candidate architectures under one protocol rather than assembling figures from separate papers.

7.6. The gap this review identifies and the study it motivates

Section 6.6 stated the absence. No row of Table 2 combines four things: several ImageNet-pretrained CNN backbones trained under one protocol, a Vision Transformer benchmark trained under the same protocol, an analysis of what the trained models attend to, and a cross-generator test. Each element exists somewhere in the table — GenImage benchmarks a ResNet-50 alongside transformer backbones (Zhu et al., 2023), Coccomini et al. (2022) vary the architecture on fixed training data, CIFAKE supplies a Grad-CAM analysis (Bird and Lotfi, 2024) — but never the four together.

The second phase of this project is planned to address that combination. None of it has been carried out; this review reports no experimental results. The plan is a transfer-learning comparison of four ImageNet-pretrained CNN backbones — ResNet50 (He et al., 2016), DenseNet121 (Huang et al., 2017), EfficientNet-B0 (Tan and Le, 2019) and VGG19 (Simonyan and Zisserman, 2015) — each in a feature-extraction and a fine-tuning configuration, benchmarked against a Vision Transformer (Dosovitskiy et al., 2021) trained under matched conditions. Every element follows from a finding above. The backbone family is the one Section 3.2 showed to be standard yet rarely compared like for like; the single matched protocol answers 7.5; the ViT is a benchmark rather than a proposed improvement, because Section 3.3 found that changing the backbone in published work usually

changes the pretraining data and objective at the same time. Grad-CAM (Selvaraju et al., 2017) will be applied to correct and incorrect predictions, because Section 3.5 showed that explanations can expose a shortcut even when they cannot name the evidence. CIFAKE (Bird and Lotfi, 2024) will be the primary dataset: at 120,000 images it is small enough to make a matched comparison across five architectures tractable on modest hardware, but it holds a single generator, so a GenImage subset (Zhu et al., 2023) will supply a cross-generator test against generators the models have not seen. That test is what would separate a result about one generator from a result about detection.

8. Conclusion

This review surveyed deep learning approaches to detecting AI-generated still images. It grouped the field by the origin of the evidence a detector uses — handcrafted and frequency-domain forensics, convolutional networks trained by transfer learning, transformer and attention-based models, frozen vision-language representations, and explainability-oriented work — then compared the datasets and benchmarks in common use, the metrics the literature reports, and 48 primary studies side by side in a single table that records for each one whether it was tested on generators absent from its training data and what limitation its own authors state.

Three findings stand out. The first is that the generalisation gap is structural rather than incidental: much of what a detector learns belongs to a particular generator's design, so a change of design removes the evidence rather than merely making it harder to read. The reported drops are large, from 99.9% accuracy on a training generator to 54.9% on an unseen one (Zhu et al., 2023), and from 92.77% on a standard benchmark to 65.77% on realistic images (Yan et al., 2025a). The second is that the strongest cross-generator results share three properties: a representation not trained on detection data, an input derived from the image rather than the image itself, or training data spanning many generators. The third is that cross-generator evaluation and explanation analysis are rarely done together — 36 of the 48 studies report the former, four the latter.

That last observation is the gap this review identifies, and Section 7.6 sets out the study planned to address it: a matched comparison of four ImageNet-pretrained CNN backbones against a Vision Transformer benchmark on CIFAKE (Bird and Lotfi, 2024), with Grad-CAM (Selvaraju et al., 2017) applied to correct and incorrect predictions and a cross-generator test on a GenImage subset (Zhu et al., 2023). That work belongs to the project's second phase. No part of it has been carried out, and this paper reports no experimental results of its own.

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